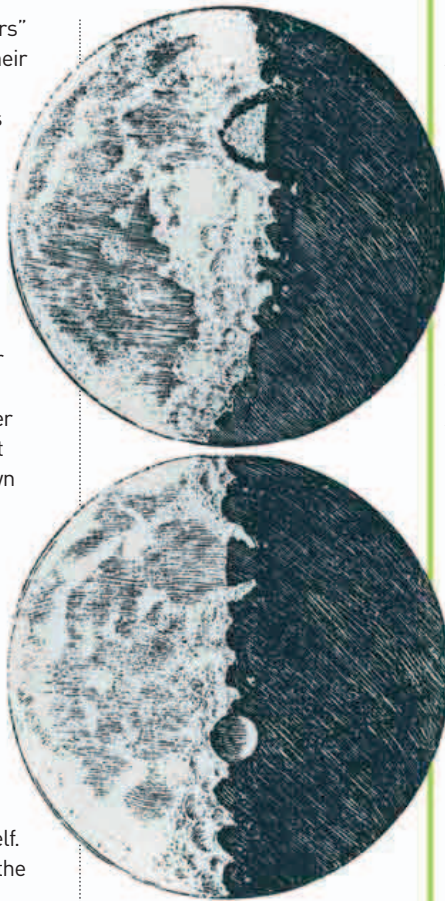


“ IN QUESTIONS OF SCIENCE, THE AUTHORITY OF A THOUSAND IS NOT WORTH THE HUMBLE REASONING OF A SINGLE INDIVIDUAL. ”

Galileo Galilei, Italian mathematician and astronomer, 1632

previously undetected “stars” close to Jupiter. However, their behavior indicated that they were, in fact, not stars but moons or satellites, orbiting the planet—a theory confirmed when one of them disappeared behind Jupiter. In further observations, he discovered a fourth satellite following a similar orbit. These **Galilean satellites**, as they were later called, were the four largest moons of Jupiter, now known as Io, Europa, Ganymede, and Callisto, their names being associated with classical myths.

The telescope also led to new discoveries elsewhere: French astronomer **Nicolas-Claude Fabri de Peiresc** (1580–1637) acquired one in 1610 and saw the Galilean satellites for himself. Later that year, he became the **first person to observe the Orion Nebula**.



Galileo's moon map

Although not the first maps of the Moon, Galileo's detailed charts were the first to show the distinctive craters and mountains on its surface.

GALILEO CONTINUED HIS ASTRONOMICAL DISCOVERIES

in 1611, describing temporary dark areas seen on the surface of the Sun—now called **sunspots**. Although he claimed to be the first to have observed these, others may have done so beforehand. The importance of their discovery lay in the fact that the periodic appearance of sunspots was yet another **challenge to the Aristotelean notion** of the perfect immutability of the heavens.

In 1611, **Kepler** published a treatise on optics, *Dioptrice*, in which he explained the **workings of the microscope** and the **refracting telescope**. He also explored the effects of using lenses of different shapes and focal lengths. He explained the workings of the Galilean telescope, with its convex and concave lenses, and also suggested a way of improving Galileo's design using two convex lenses to achieve greater magnification.

In the same year, Kepler also wrote an extraordinary “thought experiment” entitled *Somnium* (*The Dream*), which was published posthumously. In it, he

GALILEO GALILEI (1564–1642)

Born in Pisa, Italy, Galileo Galilei studied medicine and mathematics at university. In 1592, he took up professorship at Padua. His interests included astronomy and science of motion.

Galileo's scientific views were seen as heretical by the Catholic Church, and he was placed under house arrest in 1633, where he remained until his death.



described a form of interplanetary travel, and attempted to explain a model of the universe from a perspective that is not geocentric (Earth-centred).

Florentine priest and chemist, **Antonio Neri** (1576–1614) devoted much of his time to the study of glassmaking. In 1612, he published a comprehensive book *L'Arte Vetraria* (*The Art of Glass*) on the manufacture and uses of glass, which remained a standard textbook until the 19th century.

Galileo was interested not only in astronomy, but also in many other fields. In 1613, he studied

the **concept of motion**, and put forward his **principle of inertia**, which states that “a body moving on a level surface will continue in the same direction at constant speed unless disturbed.” It explained that moving objects retain their velocity unless a force, such as friction, acts upon them, a principle later important for Isaac Newton's First Law of Motion (see pp.120–21).

Galileo's telescope

Based only on vague descriptions of Lippershey's telescope, Galileo made a telescope with a combination of convex and concave lenses.



1610 Galileo identifies four of Jupiter's moons

1610 Nicolas-Claude Fabri de Peiresc discovers the Orion nebula

1611 Galileo observes and describes sunspots

1612 Antonio Neri publishes the first textbook on glassmaking, *L'Arte Vetraria*

1611 Kepler publishes *Dioptrice*, on the optics of the telescope, and the fantasy *Somnium*

1613 Galileo first describes his principle of inertia